

# Big Data Analytics Project



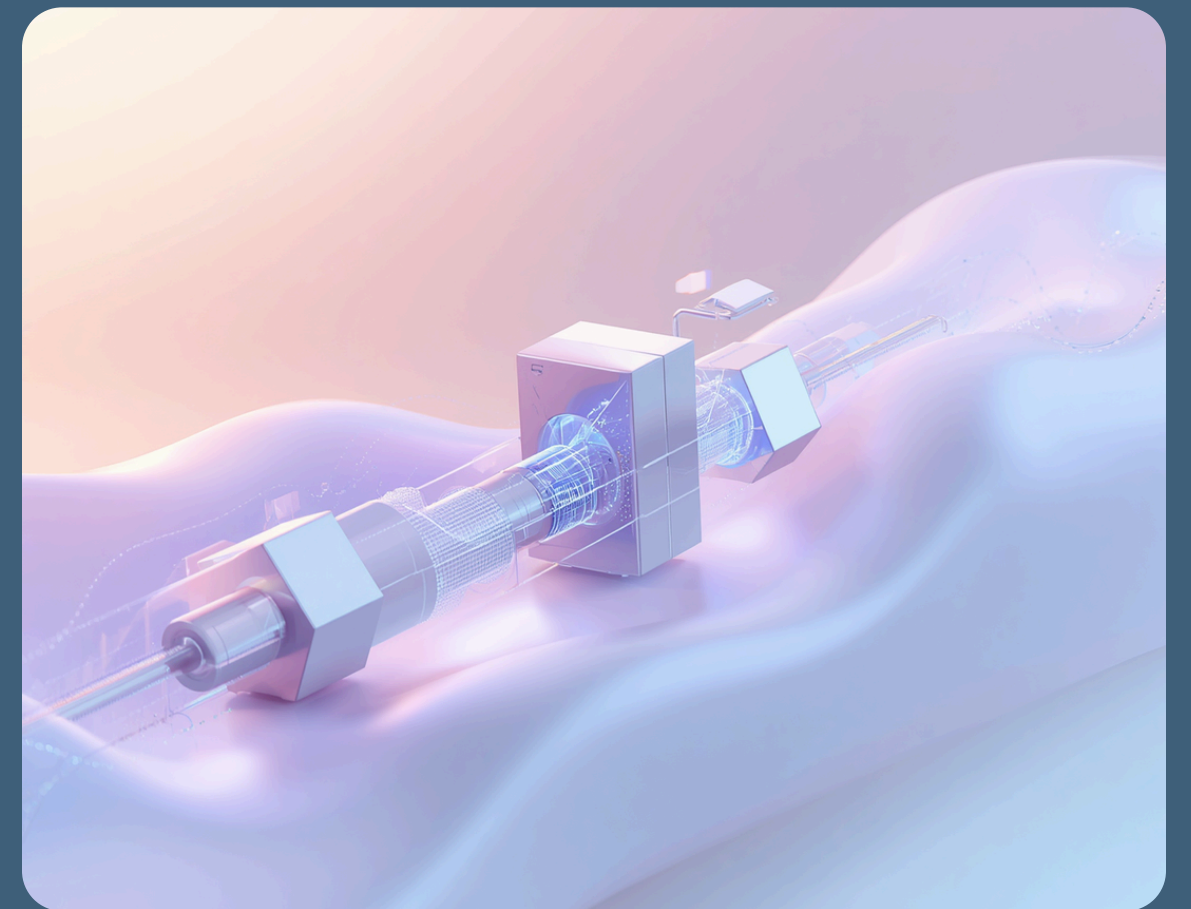
# Addressing Unstructured Data Challenges

Our project tackles the growing **massive unstructured review data** problem, overcoming scalability limitations with an automated scalable pipeline designed to efficiently process and analyze large datasets across diverse domains.

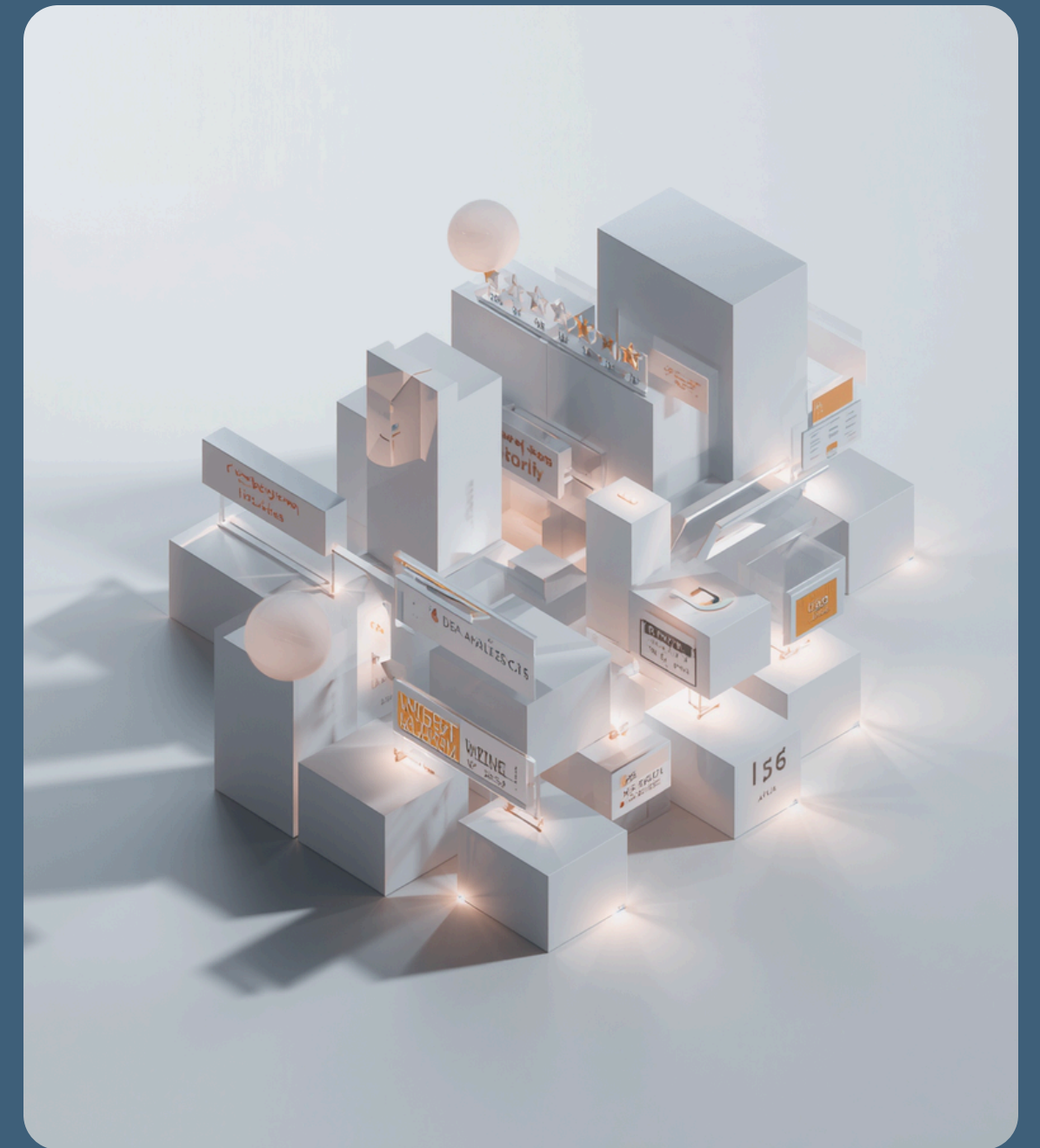


# System Architecture: End-to-End Processing

Visualizing the Complete Data Flow



# Datasets Overview: Key Domains and Purposes



# Data Cleaning Workflow Overview

This slide showcases the **data cleaning** workflow, detailing steps for handling unstructured review data and generating Parquet outputs for efficient storage and processing in our analytics pipeline.





# Sentiment Analysis Overview

Exploring Classification and Feature Engineering

This section delves into **rating-based classification** techniques, showcasing keyword feature engineering and the resulting scores in MongoDB, enhancing our understanding of user sentiments effectively.

# Product Recommendations

## Utilizing ALS for Enhanced Suggestions

### Collaborative Filtering

Collaborative filtering leverages historical user preferences to recommend products, ensuring personalized experiences. This technique effectively predicts user interests based on similar profiles, enhancing user engagement significantly.

### Matrix Factorization

Matrix factorization decomposes user-item interactions into latent factors, enabling better insights into user preferences. This method improves accuracy in recommendations and optimizes the prediction engine's performance effectively.

### Cold-Start Handling

Addressing cold-start issues involves utilizing additional data sources and algorithms for new users or items. This strategy ensures continuous recommendation quality despite limited initial information, enhancing user satisfaction.



# MongoDB & Hive Integration

**Leveraging Data for Efficient Analytics**

## Collections Overview

MongoDB collections store various datasets, enabling flexible queries and efficient retrieval. This architecture enhances data accessibility and supports dynamic analytics for user interactions and insights.

## Querying Mechanisms

Hive integration allows for SQL-like queries on large datasets stored in MongoDB. This synergy enables seamless data analysis while utilizing familiar syntax for enhanced user experience and productivity.



# GUI Dashboard Overview



The Tkinter dashboard enables **live data visualization** of sentiment scores and product recommendations, creating a user-friendly interface that facilitates real-time insights and allows for effective decision-making in data analysis.

## Overview of Technologies Used in Project

The project utilizes a diverse technology stack to ensure efficient data processing and analysis:

- **Apache Spark:** Fast data processing engine for big data analytics.
- **HDFS:** Distributed file system for storing large datasets.
- **MongoDB:** NoSQL database for flexible data storage.
- **Python:** Programming language for scripting and data manipulation.
- **Hive:** Data warehousing solution for querying and managing large datasets.
- **MLlib:** Machine learning library for scalable algorithms.
- **Tkinter:** GUI toolkit for building interactive user interfaces.
- **Java:** Additional language support for various components.





# Future Innovations

Embracing advanced technologies for  
growth